

Mapping the landscape of scientific publications

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Completed on 05/08/2024



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Table of Contents



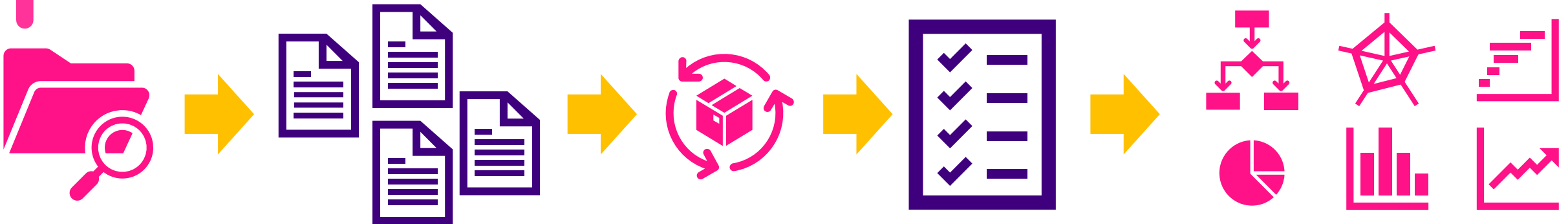
- **1/ Introduction**
 - [Why is metadata analysis important?](#)
- **2/ Dataset**
 - [Description and variables](#)
- **3/ Study design**
 - [What would be interesting to explore?](#)
 - [Experimental design](#)
 - [Tools](#)
- **4/ Data Handling**
 - [Inspection](#)
 - [Wrangling and Cleaning](#)
- **5/ Exploratory Data Analysis**
 - [Publication trend](#)
 - [Which article types are favored ?](#)
 - [In which journals are most articles published?](#) [Journal trends.](#)
 - [Is there a link between publishers and journals?](#)
 - [What is the main language used in publications?](#)
 - [Which authors are the most prolific?](#)
 - [Citation trend](#)
 - [Title length trend](#)
 - [Subject trend](#)
 - [Subject categorisation using text-mining.](#) [Cluster trends](#)
- **6/ Conclusions**
 - [What did we learn?](#)
- **7/ Future directions**
 - [What could we do next?](#)
- **8/ File Links**
 - [GitHub and Tableau Public](#)

1/ Introduction

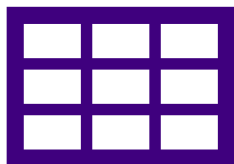
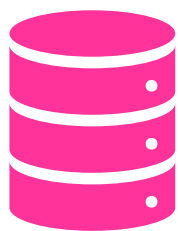
Introduction

Why is metadata analysis important?

- Metadata is defined as the information that describes and explains data in a detailed and unambiguous way.
- It provides context with details to understand the relevance of a dataset and how to use it.
- It has various applications in different research areas.
- In this project, I chose a dataset detailing metadata from scientific publications in various science fields (natural sciences, engineering and technologies, medical and health sciences, agricultural sciences, social and behavioral sciences, humanities, arts, etc...).
- I did not produce the dataset but retrieved it from AI Engineer Salman alsheikh via Kaggle (source on see dataset slide).
- Analysing this data will help me understand publication trends and topics of interest.



2/ Dataset



Dataset

Source: <https://www.kaggle.com/datasets/s4lman/scientific-articles-metadata-dataset>

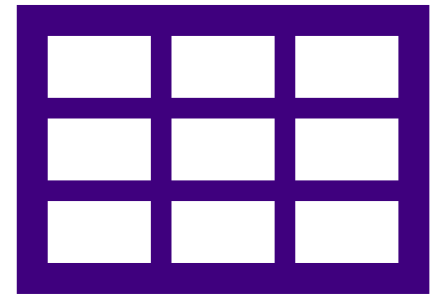
Description: This dataset contains detailed metadata on scientific research articles, lending itself to bibliometric analysis, and tracking research trends via Exploratory Data Analysis (EDA).

Raw Dataset dimensions: 120,000 rows x 16 columns

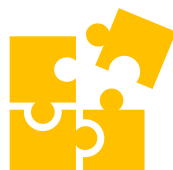
Dataset variables:

- **DOI:** Digital Object Identifier, the unique ID of a publication
- **title:** Title of the article
- **author:** Name of first author
- **issued:** Date of publication
- **abstract:** Abstract of the article
- **publisher:** Publisher of the scientific journal
- **container-title:** Name of the scientific journal
- **volume:** Volume number of the scientific journal
- **issue:** Issue number of the scientific journal
- **URL:** URL of DOI
- **score:** ?
- **references-count:** Number of references used in the article
- **language:** Language used in the article (2 letters abbreviations)
- **subject:** Topic of the article
- **type:** Type of article
- **indexed:** Date of article indexing

kaggle



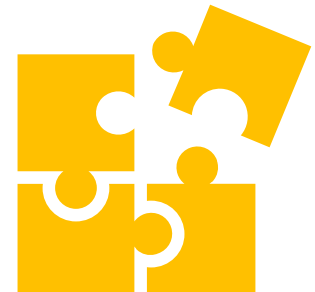
3/ Study design



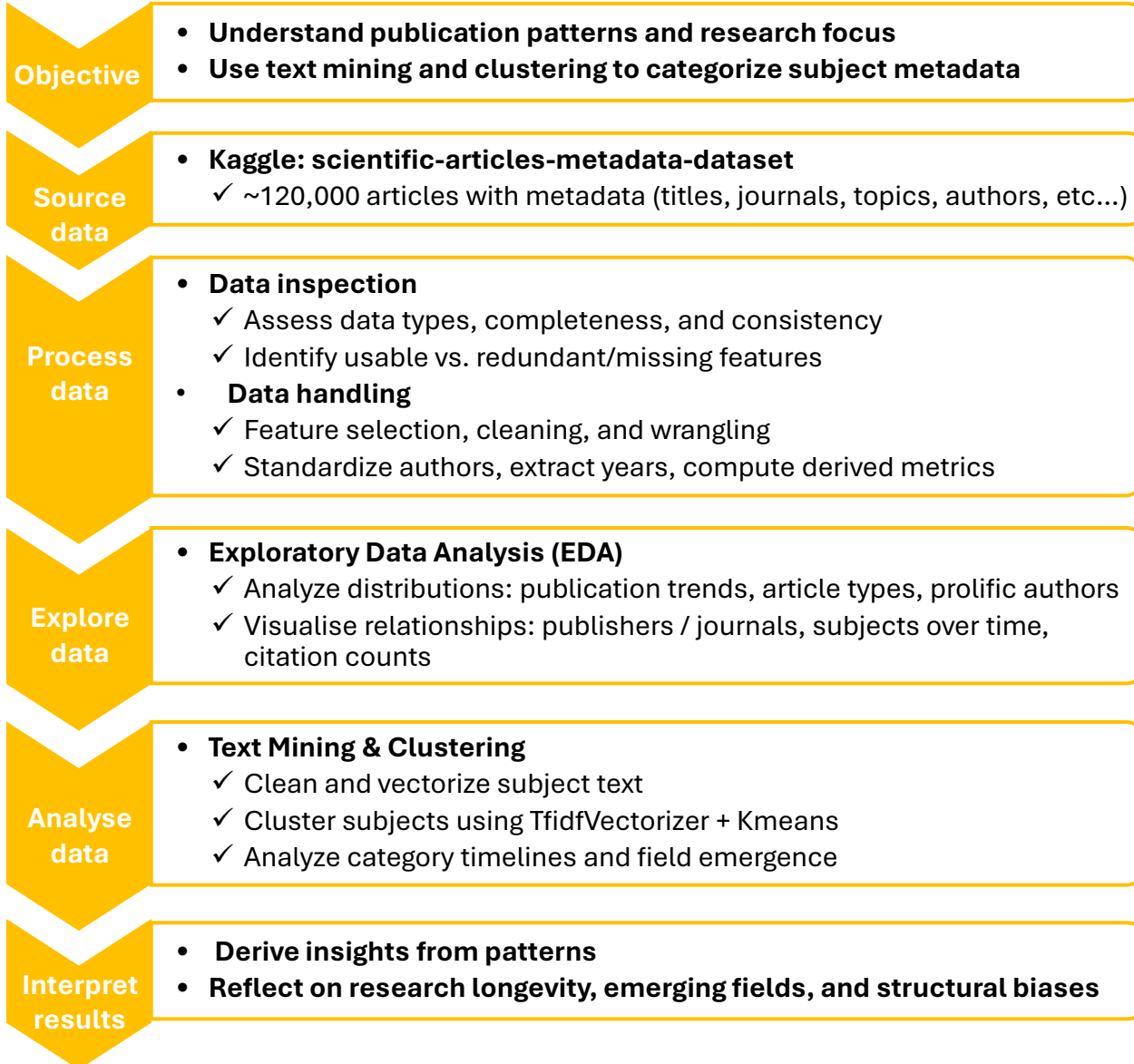
3.1/ Questions

What would be interesting to explore?

- Publication rate over time
- Successful publishers and favored journals
- Prolific authors
- Number of references and length of title over time
- Preferred publication types
- Languages used
- Most popular subjects



3.1/ Experimental design



Principle

The analysis simulates how bibliometric metadata can be harnessed to map the evolution of science.

Emphasis is placed on interpretable trends, not predictions.

3.2/ Tools

Data handling

python

Excel

python
pandas

IP[y]:
python
iPython

python
NumPy

/(reg) /ex/
python
RegEx

Data analysis

python
scikit-learn

python
NLTK

python
statsmodels

python
SciPy

python
squarify

python
tqdm

python
GENSIM

Data visualisation

python
matplotlib

python
seaborn

python
plotly

python
wordcloud

Tableau

Reporting & Publishing

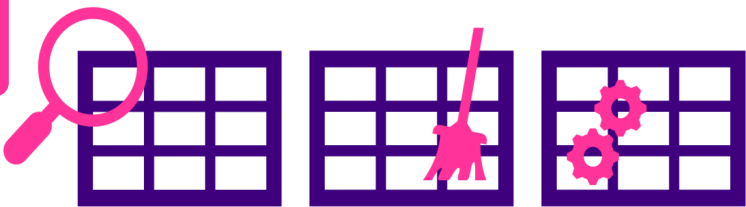
lab
JupyterLab

Anaconda
Navigator

PowerPoint

GitHub

4/ Data Handling



4.1/ Data Inspection

Data assessment and feature selection (KEEP / REMOVE):

- DOI: looks fine, keep as it is. KEEP
- title: few missing values, capitals, special characters, non sensical words, incomplete title, date instead of title etc... KEEP
- author: many missing values, inconsistent format (separators, capitalisation, first name initialed, etc...). KEEP
- issued: the dates formats are varied, needs to be uniformized, extract Years, Months. Convert 'None' to missing values. KEEP
- abstract: mostly missing values, don't use. REMOVE
- publisher: looks fine, keep as it is. KEEP
- container-title: few missing values, otherwise it looks fine, keep as it is, rename header as 'journal'. KEEP
- volume: missing values, inconsistent format (numbers, letters, dates), unnecessary field. REMOVE
- issue: missing values, inconsistent format (numbers, letters, dates), maybe I don't need to use them. REMOVE
- URL: redundant information with DOI, can be removed. REMOVE
- score: always 0, can be removed. REMOVE
- references-count: looks fine (integers), keep as it is, only numerical variable. KEEP
- language: missing values but otherwise it looks fine, keep as it is. KEEP
- subject: missing values but otherwise it looks fine, keep as it is. KEEP
- type: looks fine, keep as it is. KEEP
- indexed: looks fine, keep as it is (dates from 2022-2023). Not sure how relevant it is. REMOVE

All the variables are categorical, except for 'references-count' which is numerical (integer).

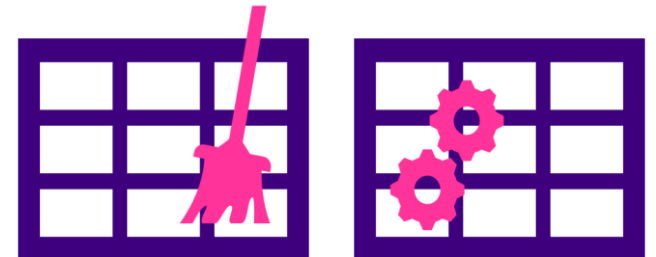


4.2/ Data Cleaning and Wrangling

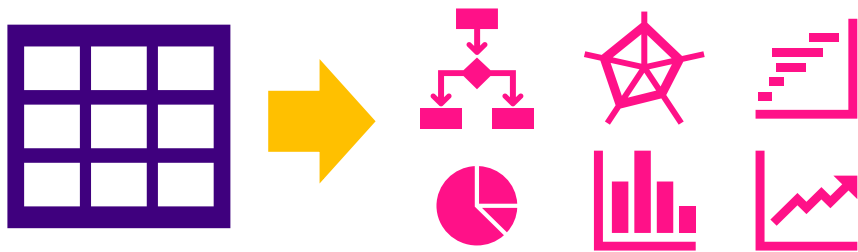
Data wrangling and cleaning:

- **author**: the last names were extracted using `str.extract()` into new column '**author_lastname**'. Must keep '**author**' as well.
- **issued**: the dates were coerced into datetime formats using `pd.datetime()` into new column '**publication_date**'; the year was extracted using `str.extract()` into new column '**publication_year**'. The '**issued**' column was removed.
- **container-title**: renamed header as '**journal**'
- **title**: it required a multistep approach:
 - remove rows with no authors as they do not correspond to real scientific articles; this eliminates title like dates and numbers as well as some of the empty titles.
 - copy the titles into a new column '**better_title**'
 - using `str.lower()` convert '**better_title**' to lower case
 - using `str.l replace('[^a-zA-Z0-9\s]', '')` remove special characters
 - using `str.len()` compute title length into new column '**title_length**'
 - using `fillna('Unknown', inplace=True)`, replace missing values with unknown

Clean Dataset dimensions: 103,282 rows x 13 columns



5/ Exploratory Data Analysis (EDA)



5.1/ Publication Trend

How do publication numbers track over time?

This dataset ranges from 1891 to 2014 with some gaps.

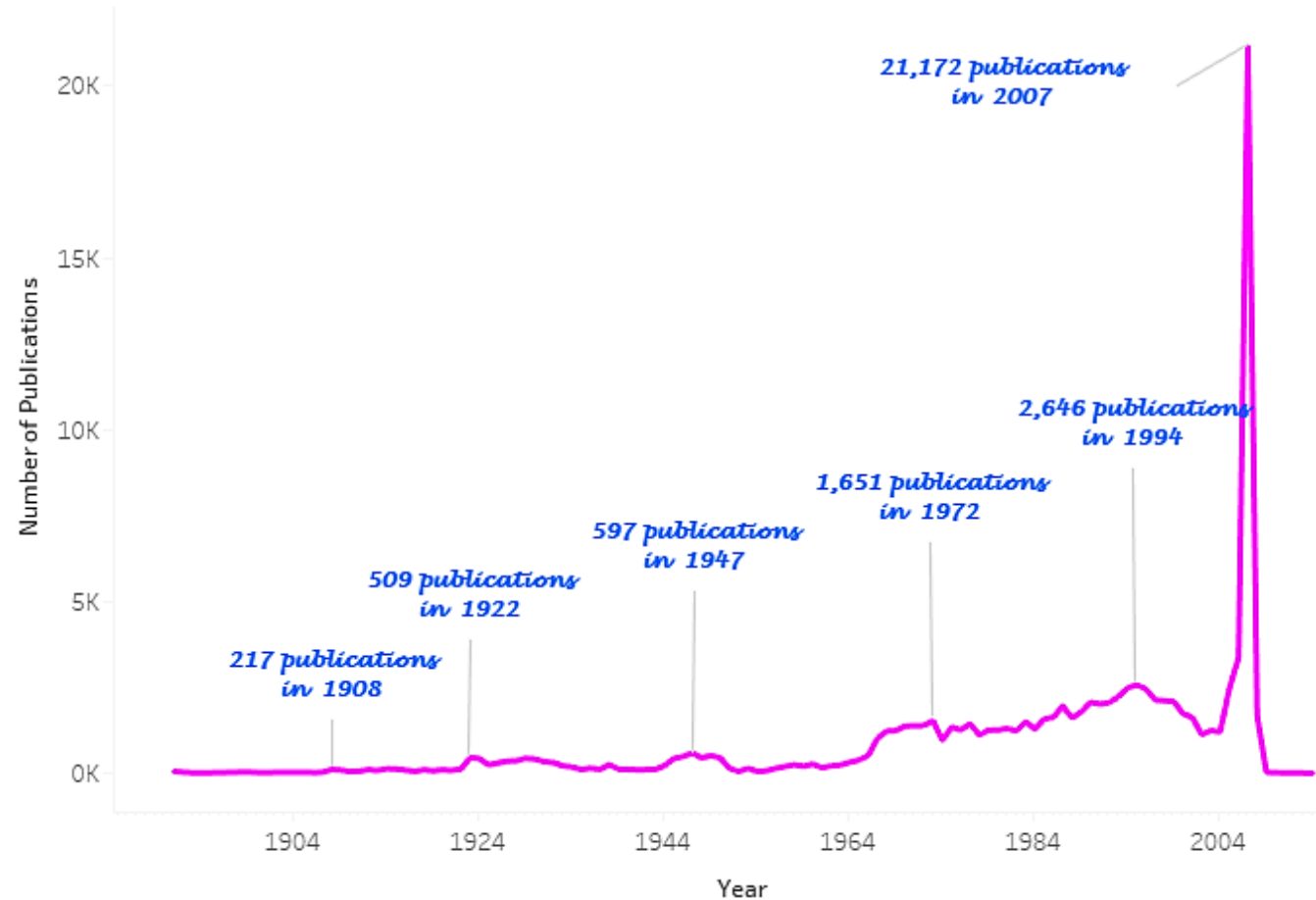
The yearly rate of publication is fairly constant from 1891 to 1960, after which it gradually increases to reach 2,500 publications per year in the mid-90's.

There is a huge spike in 2007 featuring 21,153 publications. Is it a dataset bias or did something happen to explain such spike?

There is a gap between 2010 and 2014.

Years 2009, 2010, and 2014 list only 22, 12, and 1 publications, respectively. Perhaps, those years are outliers and can be ignored.

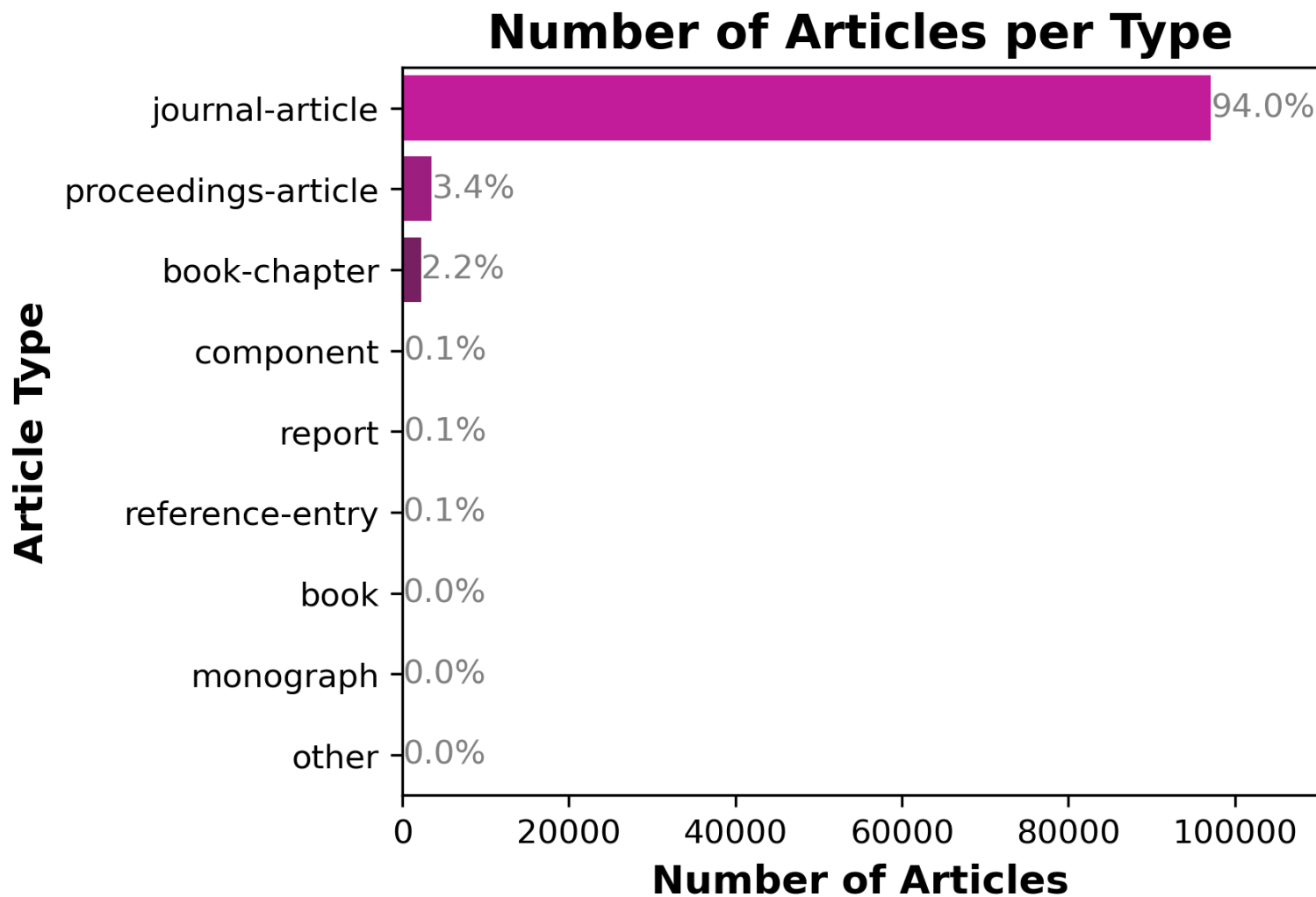
Number of Publications From 1891 to 2014



5.2/ Article Types

Which article types are favored?

Among the 9 types of article, the most favored by far are journal article (94%,) followed by proceeding articles (3%), and book chapters (2%).

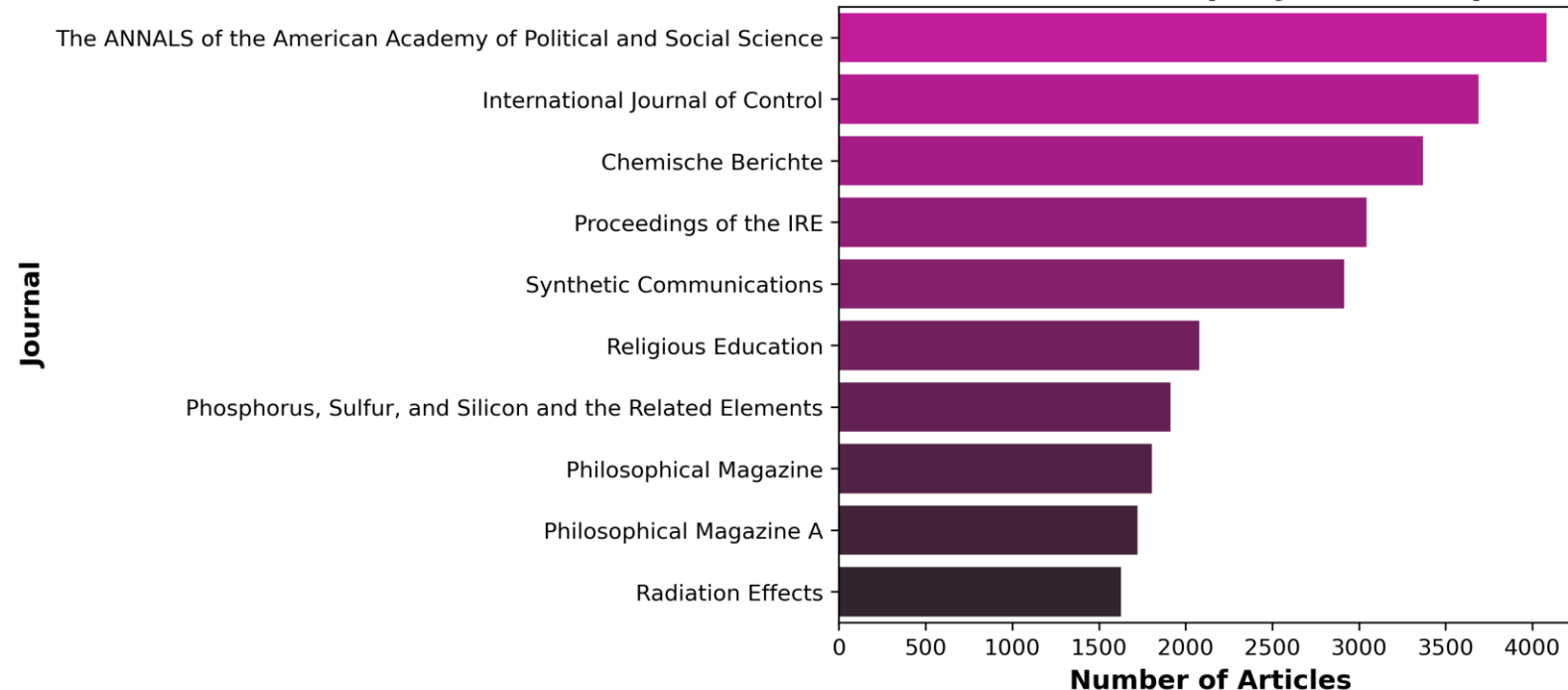


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The top journal deals with sociology, social and political sciences.



Number of Articles per Journal (Top 10)



5.4/ Journal Trend

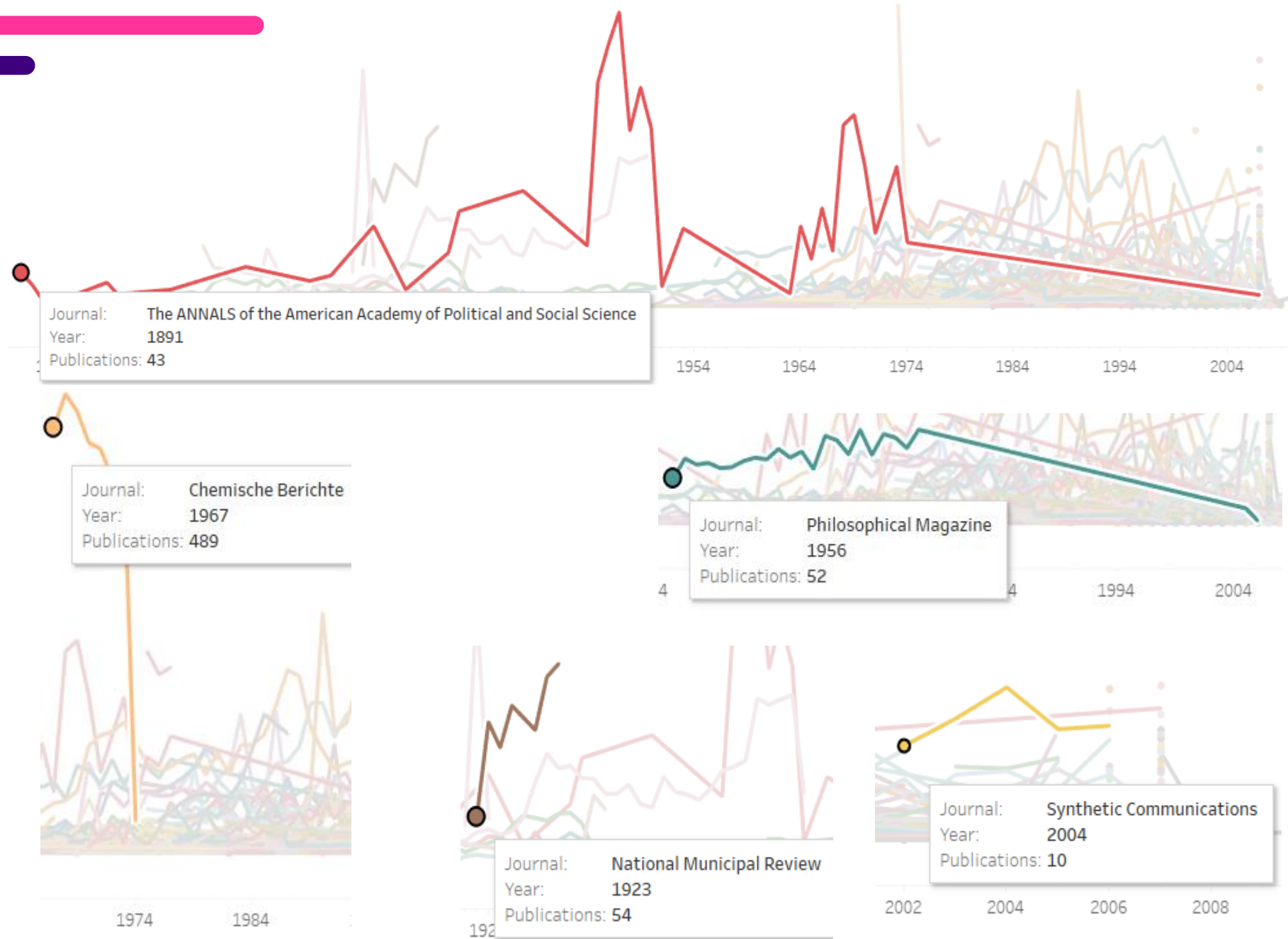
How do journals evolve over time?

Some journals span the whole timeline (e.g. “The ANNALS of the American Academy of Political and Social Science”).

Others have appeared later and are durable (e.g. “Philosophical Magazine”).

Some are short-lived (e.g. “Chische Berichte” 1967-1974, “National Municipal Review” 1923-1930).

The journal “Synthetic Communications” emerged in 2004.



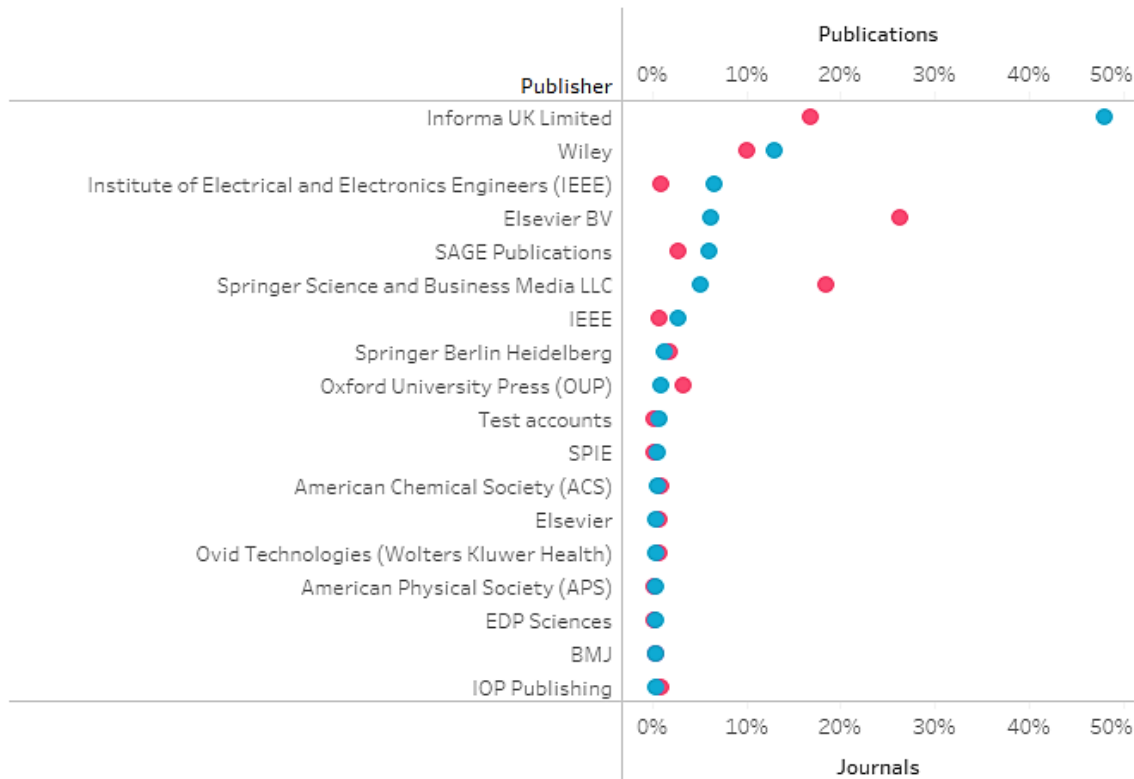
5.5/ Publishers

Is there a relationship between publisher and journals?

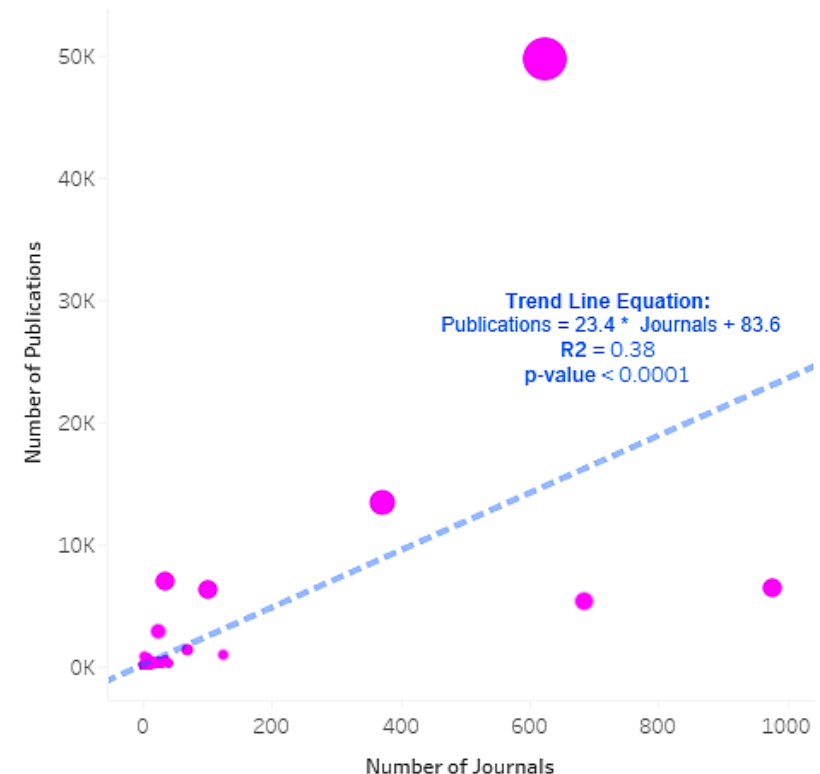
Among the 194 publishers, the most successful are Informa UK Limited (49628 publications, 48%) and Wiley (13324 publications, 13%). Publishers that host the largest number of journals are Elsevier BV (977 journals), Springer Science and Business Media LLC (685 journals), Informa UK Limited (623 journals), and Wiley (372 journals).

There is a positive relationship ($R^2=0.4$) between number of publications and number of journals per publisher.

Number of Journals and Publications per Publisher



Number of Publications and Journals per Publisher

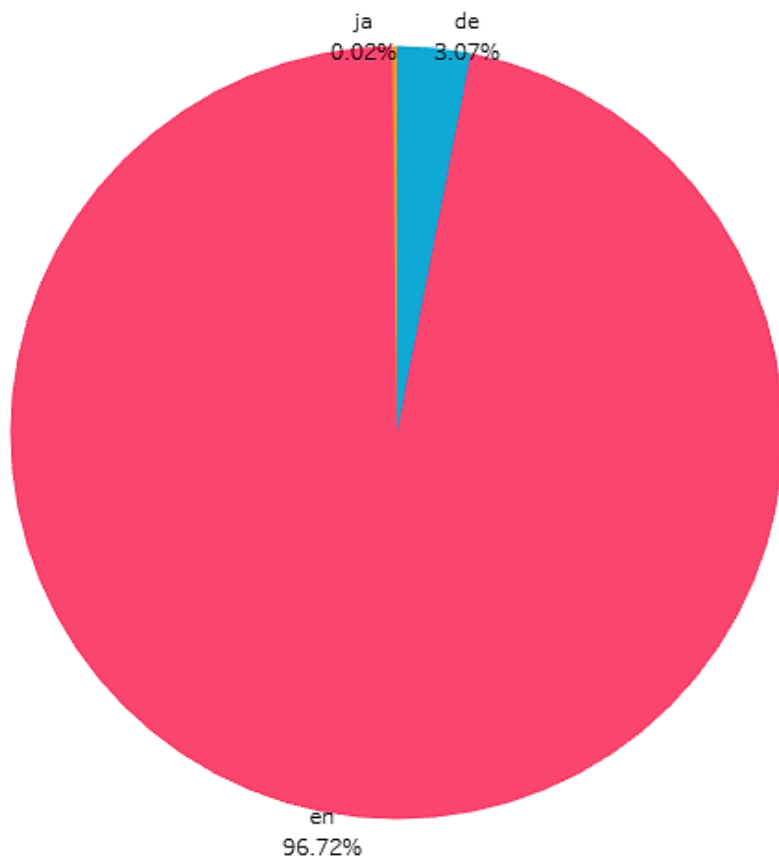


5.6/ Main Languages

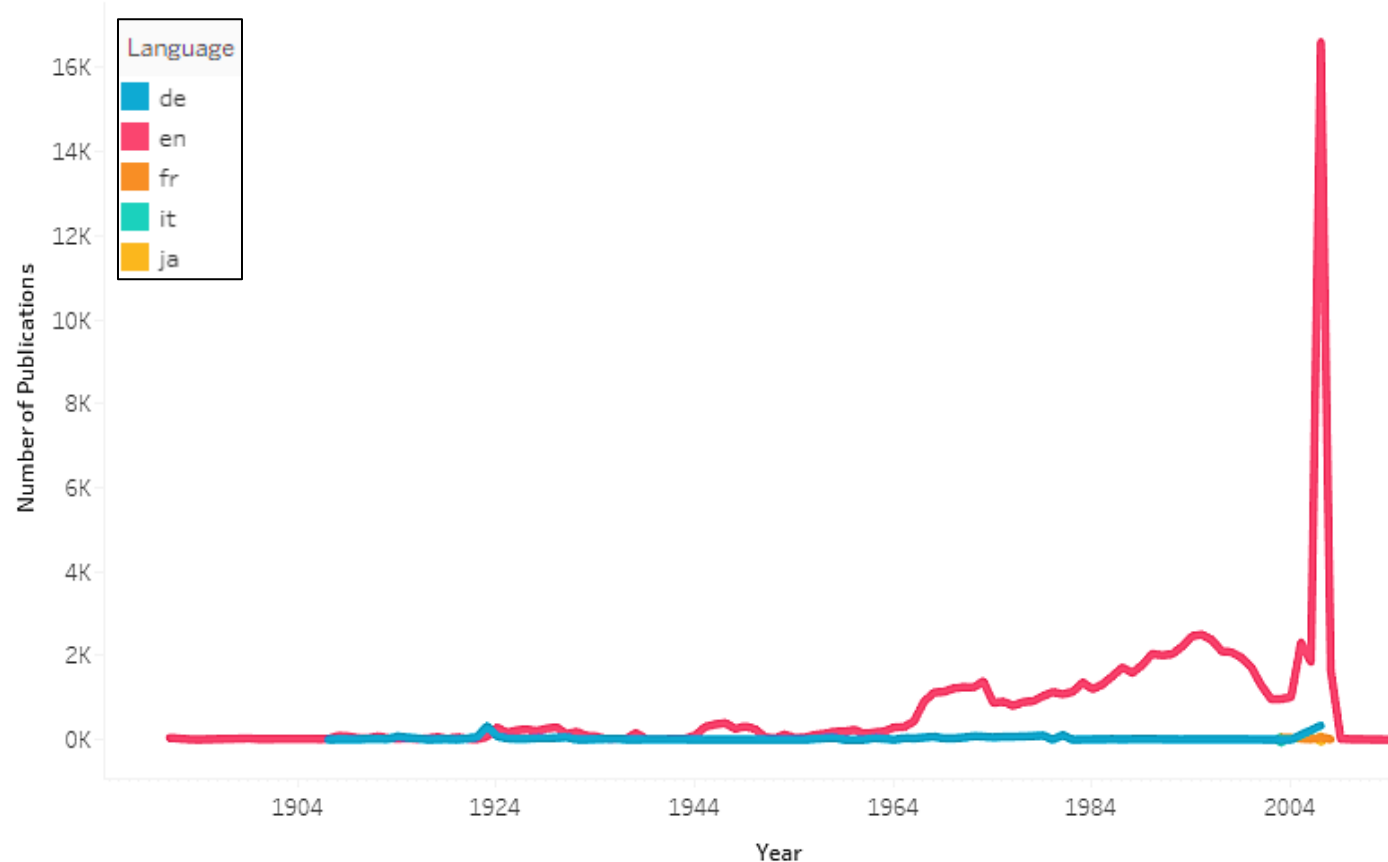
Which languages are used?

A total of 5 languages are listed: en, de, fr, it, ja. English is overwhelmingly (97%) the language used for publication purpose, followed by Deutsch (3%). French is only used from 2003-2008, Italian in 2003 and Japanese in 2007.

Percentage of Publications per Language



Publications per Language from 1891 to 2014



5.7/ Prolific Authors

Which authors are the most prolific?

Among the 81,743 first authors, the most prolific ones are Ferdinand Bohlmann a chemist interested in inorganic chemistry with 124 publications, followed by D.A. Michalopoulos interested in Computer Science with 68 publications, John Bauer a municipal public servant with 60 publications, and D.J. Evans a mathematician with 54 publications.

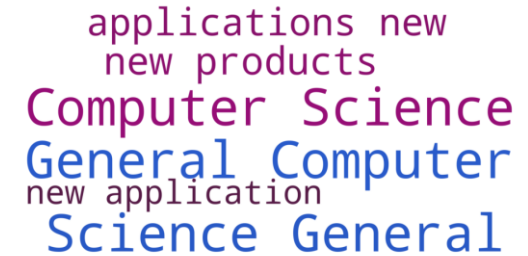
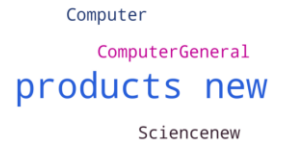
Number of Articles per Author (top 10)

John Bauer 60	Hubert Schmidbaur 44
Russell Forbes 47	T.R. Gilliland 45
D.A. Michalopoulos 68	C. Richard Conti 53
Ferdinand Bohlmann 124	Hans Paulsen 50
D. J. Evans 54	L.W. Austin 53

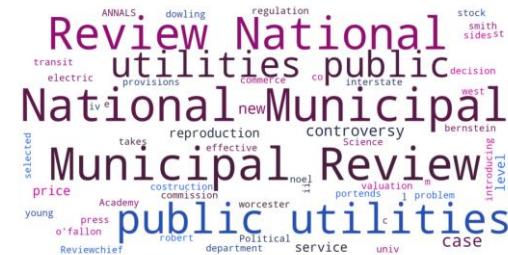
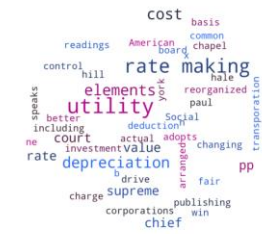
1. Ferdinand Bohlmann



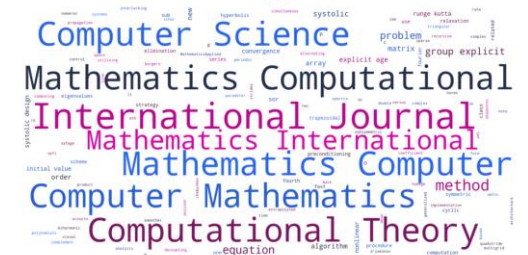
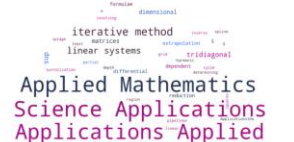
2. D.A. Michalopoulos



3. John Bauer



4. D.J. Evans

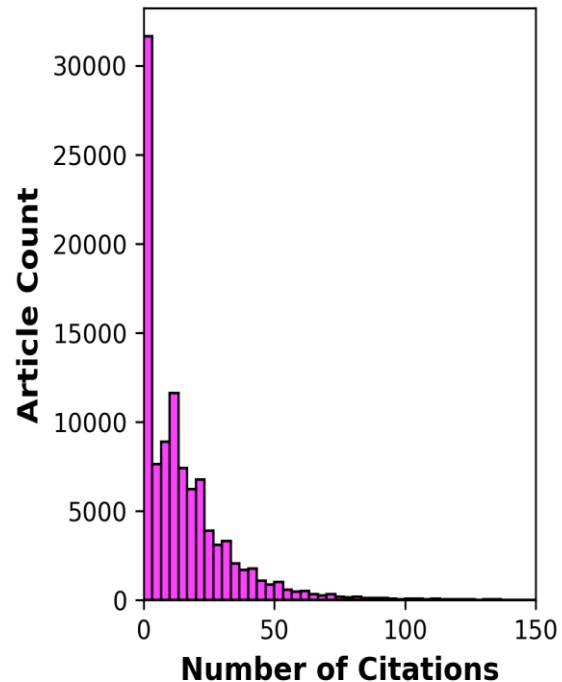


5.8/ Citations

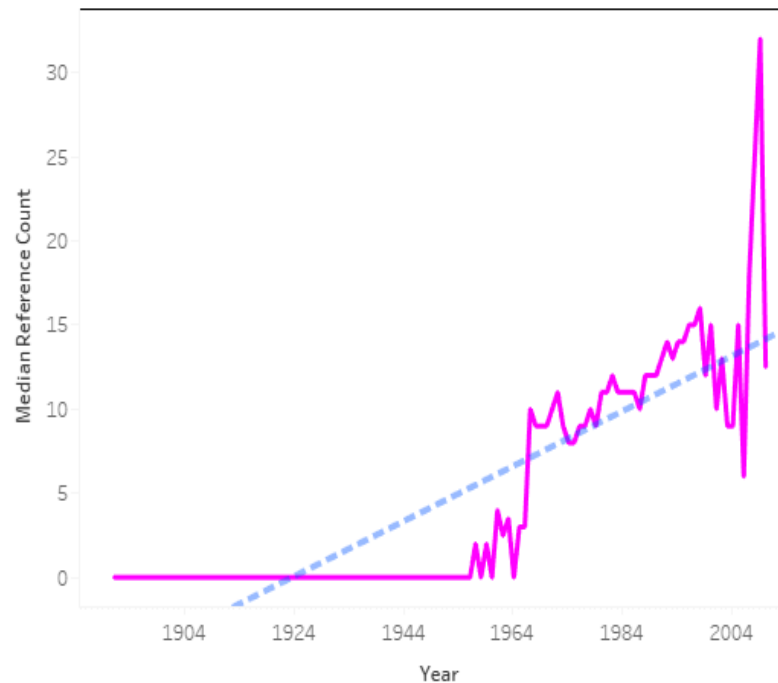
How do citation numbers track over time?

The distribution is highly right-skewed; 75% articles have < 22 references, with a few outliers featuring > 200 citations. The average number of references per article was almost non-existent from 1891-1955, bar 2 spikes in 1913-1916 (33 citations in 1914) and 1934 (5 citations). It steadily increased from 1956 to 2010 with a peak at 35 in 2009. Reference entries exhibits the widest range of citation numbers. Book chapters typically list more references than other publications.

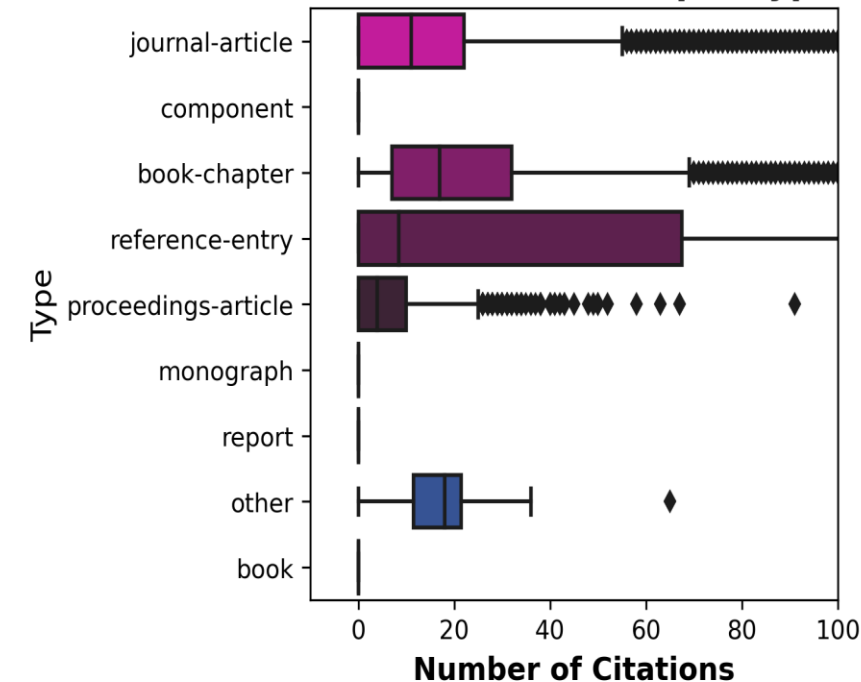
Distribution of citation numbers



Median Citation Number from 1891 to 2014



Distribution of citation numbers per type

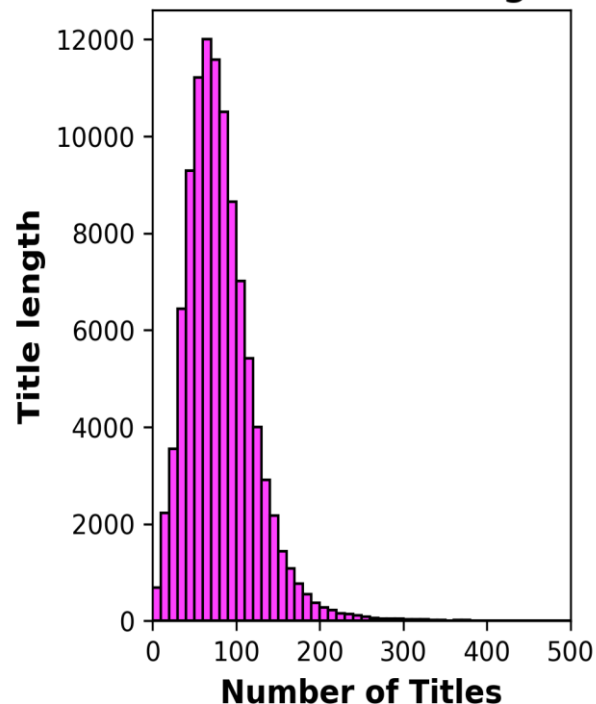


5.9/ Title lengths

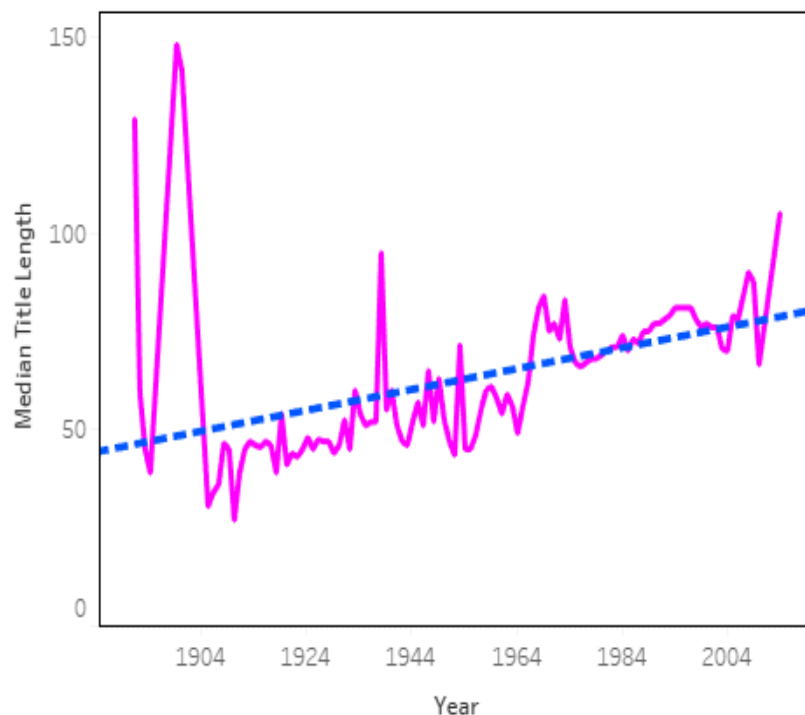
How do title lengths track over time?

The distribution is right-skewed; 75% articles have titles with < 101 characters, with a few outliers featuring > 400 characters. The average length of title is 80. While fluctuating, title lengths steadily increase from the beginning of the 20th century to 2014. Journal article entries exhibits the widest range of title length. Conversely reference entries and book feature the shortest titles.

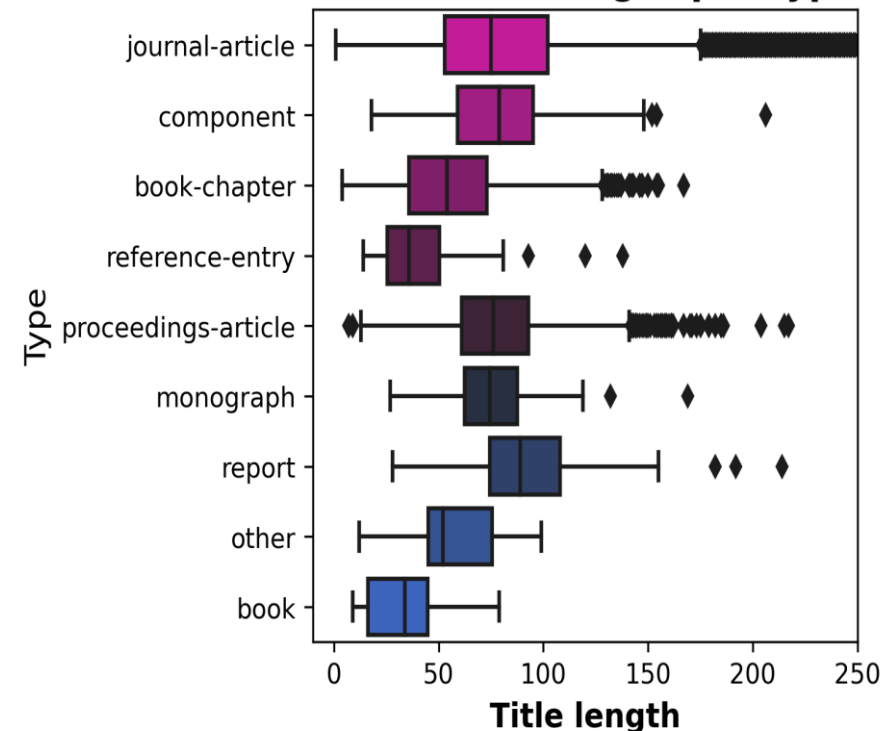
Distribution of title length



Median Title Length from 1891 to 2014



Distribution of title length per type



[illegible]

Subject

as matter physics.

Subject	Number of Articles
General Social Sciences, Sociology and Political Science	4500
Computer Science Applications, Control and Systems Engineering	3750
Electrical and Electronic Engineering	3750
General Engineering	3500
Inorganic Chemistry	3400
Organic Chemistry	3000
Religious studies, Education	2000
Inorganic Chemistry, Organic Chemistry, Biochemistry	1900
General Medicine	1800
miscellaneous), Condensed Matter Physics, General Materials Science, Electronic, Optical and Magnetic Materials	1750
General Physics and Astronomy, General Chemical Engineering	1500
Education	1375
Applied Mathematics, Computational Theory and Mathematics, Computer Science Applications	1250
Spectroscopy, Atomic and Molecular Physics, and Optics, Analytical Chemistry	1250
Condensed Matter Physics, General Materials Science, General Chemistry	1125
Filtration and Separation, Process Chemistry and Technology, General Chemical Engineering, General Chemistry	1000
Genetics, Biochemistry	1000
Software	1000
Condensed Matter Physics	900
Inorganic Chemistry, Inorganic Chemistry, Physical and Theoretical Chemistry	900

5.11/ Subject Trend

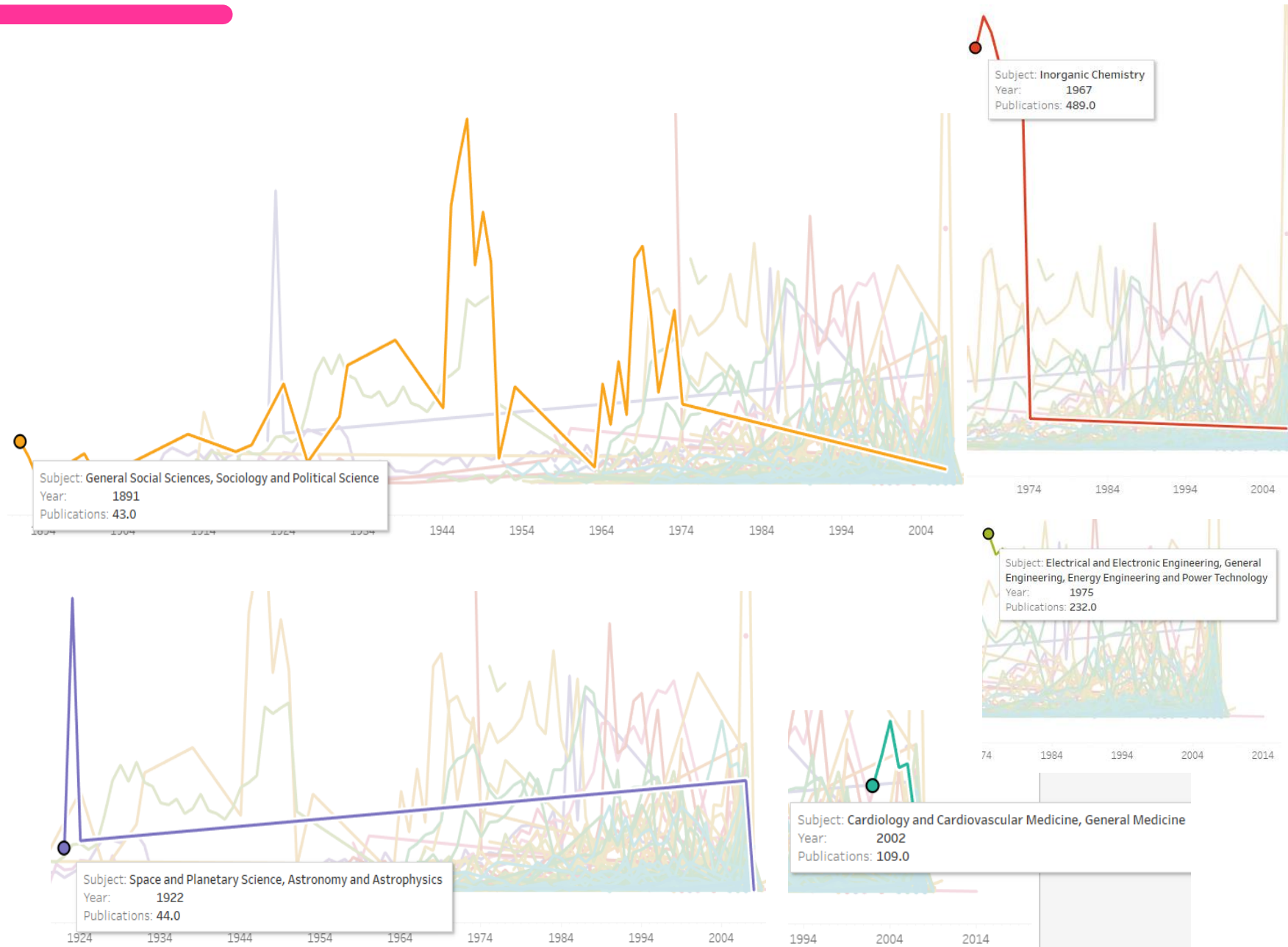
How do subjects evolve over time?

Some subjects span the whole timeline (e.g. “General Social Sciences, Sociology and Political Science”).

Others have appeared later and are durable (e.g. “Space and Planetary Science, Astronomy and Astrophysics” from 1922, “Inorganic Chemistry” from 1967).

Some are short-lived (e.g. “Electrical and Electronic Engineering, General Engineering, Energy Engineering and Power Technology” 1975-1977).

The subject “Cardiology and Cardiovascular Medicine, General Medicine” emerged in 2002.

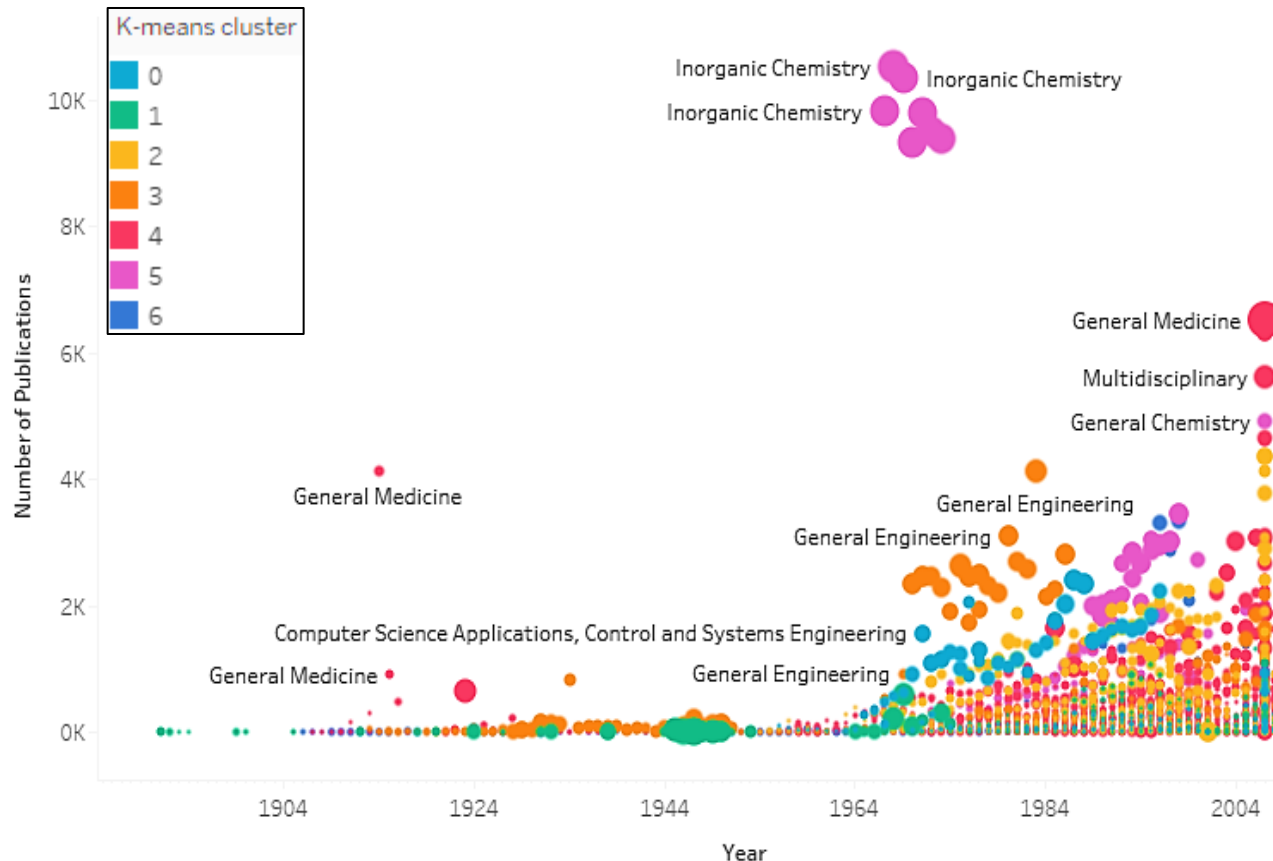


5.13/ Subject Categories - k means clusters

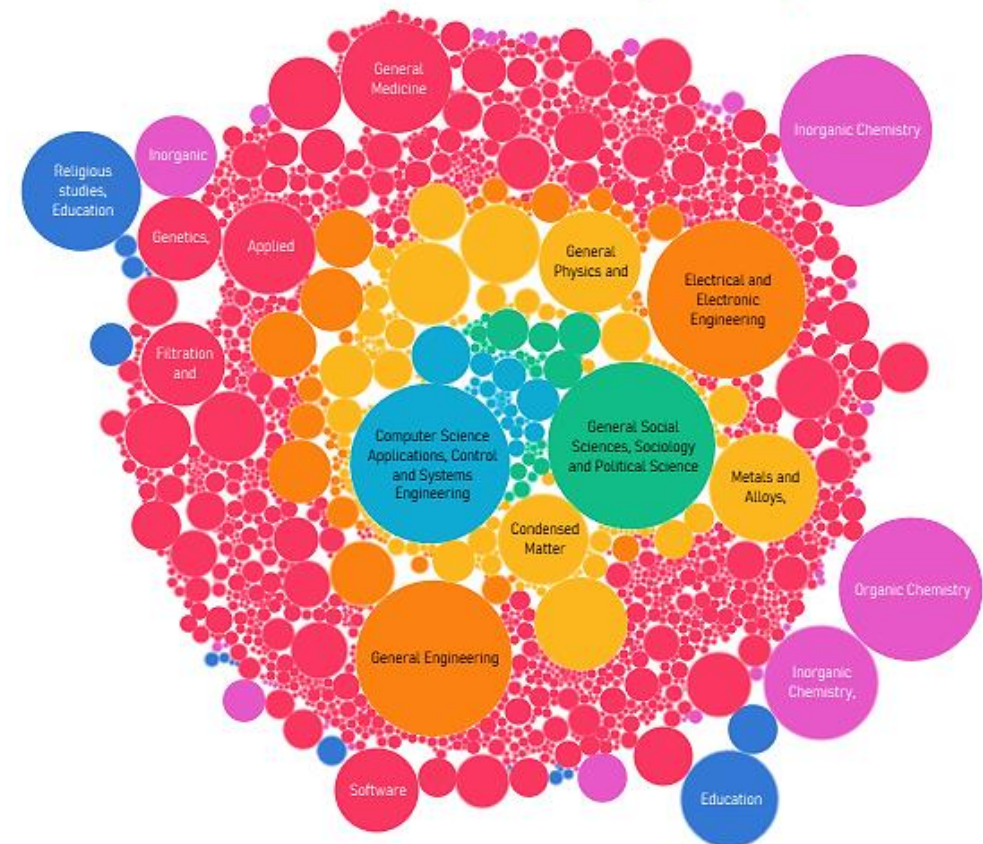
What are some of k-means cluster trends?

Having less categories (7 instead of 1265) helps visualize this complex data. A timeline shows that chemistry (K-means cluster 5) became hugely popular in the 70's but were less fashionable later on. Computer science (K-means cluster 0) surged in the 60's, with increased interest until the late 90's, coincidentally with general engineering (K-means cluster 3).

Number of Publications per Subject



Number of Publications per Subject

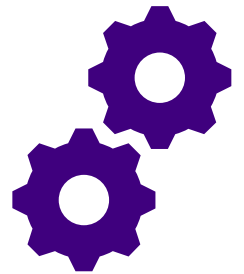
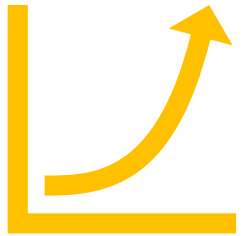
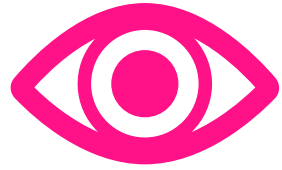


6/ Conclusions

Conclusions

What did we learn?

- Yearly rate of publication gradually increases from the 60's onward to reach 2,500 publications per year in the mid-90's
- 2007 featured many more publications than other years. Is it a dataset bias or did something happen to explain such spike?
- Most publications are journal article (94%,) followed by proceeding articles (3%), and book chapters (2%)
- The popularity and longevity of journals vary immensely
- Most successful publishers host a larger number of journals
- We covered examples of most prolific authors interested in different academic fields
- Publications are overwhelmingly written in English
- The number of citations as well as the title length are both on the rise
- The most popular subjects deal with social sciences, computer science, electrical and general engineering as well as chemistry
- Some subjects are long-lasting while others are short-lived or emerging
- As no broad subject categories were available, we made our own using text-mining and clustering which helped further analyse trends

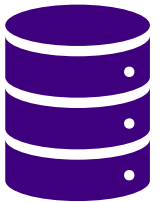


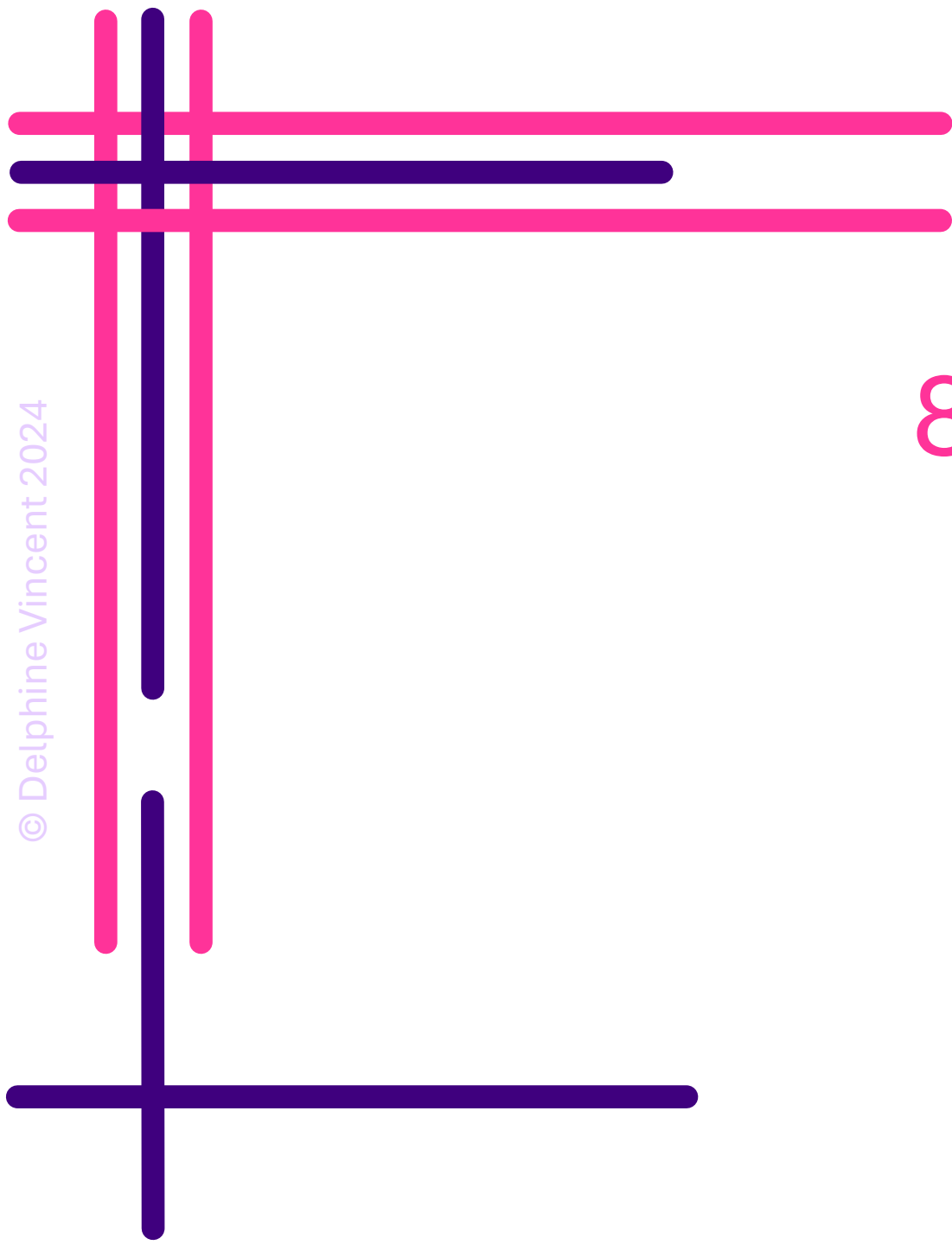
7/ Perspectives

Perspectives

What could we do next?

- Create more sophisticated visualization (e.g. Sankey or chord diagrams) to better explore relationship across variables
- Refine the text-mining models using other LNP methods
- Fetch more metadata:
 - co-authors' names to establish scientist networks
 - UNESCO academics field of knowledge to properly categorise the data
 - More numerical variables (how many times the publication was cited, submission/acceptation/publication dates to track how fast reports are processed and published)
 - Keywords for easier text-mining analysis
 - All the abstracts for more in-depth text mining
 - More detailed publication types for articles (reviews, short communication, technical note, etc...)
 - Authors' geographical area (country, region, town) to map where research hotspots are





8/ Links

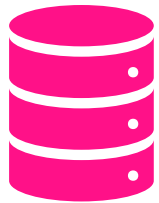


Links to Files



The following files are publicly available:

- Raw dataset: scientific_papers_metadata_20240709.csv (GitHub)
- Python code: Scientific_Articles_Metadata_dlf2024.ipynb (GitHub and Kaggle)
- Report: scientific_papers_metadata_dlf2024.pdf (GitHub)
- Tableau data visualization: scientific_papers_metadata_dlf2024.twb (GitHub and Tableau Public)



Repositories:

- GitHub: [https://github.com/dlf2024/Python Tableau scientific papers metadata](https://github.com/dlf2024/Python_Tableau_scientific_papers_metadata)
- Tableau Public:
https://public.tableau.com/app/profile/delphine.vincent/viz/scientific_papers_metadata_dlf2024/Dashboard_publications?publish=yes
- Kaggle (private): <https://www.kaggle.com/code/dlf2024/scientific-articles-metadata-dlf2024>

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